

ORGANIC CHEMISTRY – ALKENES & THEIR REACTIONS

KEY VOCABULARY

Alkene: an unsaturated hydrocarbon, formula C_nH_{2n} .

Functional group: the $C=C$ bond — where reactions happen.

Addition reaction: atoms add across the $C=C$ double bond.

Hydrogenation: adding hydrogen across $C=C$ to form an alkane.

COMMON MISCONCEPTIONS

~~Alkenes and alkanes have the same formula.~~

✓ Alkenes are C_nH_{2n} ; alkanes C_nH_{2n+2} .

~~The whole molecule reacts.~~

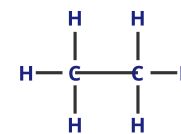
✓ Reactions happen at the reactive $C=C$ group.

~~Alkenes do not decolourise bromine water.~~

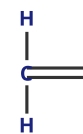
✓ They DO — the $C=C$ reacts with bromine.

ETHENE (A $C=C$ BOND)

ETHANE (alkane)

 C_2H_6 · saturated

ETHENE (alkene)

 C_2H_4 · unsaturated ($C=C$)

1 GENERAL FORMULA

Give the **general formula** of an alkene. [1 mark]

2 FUNCTIONAL GROUP

Name the functional group that makes alkenes reactive. [1 mark]

3 ADD BROMINE

Explain what happens when ethene reacts with bromine. [2 marks]

4 HYDROGENATION

Name the product when ethene reacts with hydrogen. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Alkenes react in ____ reactions because the ____ bond can open.

addition

double

single

substitution

6 STEAM + ALKENE

Name the product when ethene reacts with steam. [1 mark]